

A **35-year-old premenopausal female** with recently diagnosed breast cancer presents today for recommendations regarding treatment. Her oncologic history is as follows:

1. August 15, 2016: patient noticed a palpable right breast mass after breastfeeding.
2. September 2, 2016: patient sought medical attention for the breast mass. Diagnostic mammogram revealed a 3.5 cm right breast mass, suspicious for malignancy. Axillary ultrasound revealed two suspicious axillary lymph nodes, concerning for metastatic involvement.
3. September 5, 2016: breast MRI revealed a right breast mass measuring 3.8 cm, along with 3 pathological appearing lymph nodes.
4. September 6, 2016: Ultrasound-guided **biopsy of the right breast** reveals an **invasive ductal carcinoma**, Nottingham **grade 3** (of 3), estrogen receptor (ER) positive (**ER >75%**), progesterone receptor (PR) positive (**PR 1-10%**), human epidermal growth receptor 2 (**HER-2**) **positive** (immunohistochemistry 2+, fluorescence in situ hybridization (FISH) positive for amplification. **Biopsy of right axillary lymph node** reveals evidence for **metastatic** adenocarcinoma, consistent with breast primary.
5. September 8, 2016: initial visit with Medical Oncology. Neoadjuvant chemotherapy followed by surgery and adjuvant breast radiotherapy is recommended.
6. September 15, 2016: patient starts **neoadjuvant chemotherapy with paclitaxel, trastuzumab, and pertuzumab**.
7. January 30, 2017: patient starts dose-dense **doxorubicin, cyclophosphamide** with neulasta support.
8. February 2, 2017: comprehensive hereditary breast cancer panel reveals a germline pathogenic **BRCA1 mutation**, confirming a hereditary breast cancer predisposition syndrome.
9. April 20, 2017: patient undergoes **bilateral mastectomy** with right axillary sentinel lymph node biopsy. Breast pathology specimen reveals a **residual 0.5 cm** right breast carcinoma, **43% cellularity**, with evidence of treatment effect, and **1 out of 23 lymph nodes with** evidence of **metastatic disease**. Left breast shows only benign parenchyma.
10. May 15, 2017: patient initiates **adjuvant trastuzumab** with plans to continue 12 months of therapy along with **tamoxifen**.
11. July 3, 2017: **bilateral salpingo-oophorectomy** for prophylactic purposes, given BRCA mutation.
12. September 23, 2017: patient develops severe headaches and nausea. MRI brain ordered.
13. September 25, 2017: Brain MRI reveals **2 dominant brain metastases**.
14. September 26, 2017: Neurosurgery consultation. Recommendation is for gamma knife surgery of two major lesions.
15. September 30, 2017: patient undergoes **gamma knife surgery**.
16. October 15, 2017: systemic therapy changed to **ado-trastuzumab emtansine**, with discontinuation of the previous plan of completing 12 months of adjuvant trastuzumab.
17. February 20, 2018: evidence of **liver metastasis**. Ado-trastuzumab emtansine discontinued, and systemic treatment changed to **capecitabine/lapatinib**. Request for liver biopsy.
18. February 25, 2018: liver biopsy reveals metastatic breast cancer, positive for PDL-1.
19. March 1, 2018: progressive disease in the liver and bones. Capecitabine/lapatinib stopped. Patient starts **experimental pembrolizumab**, as part of a clinical trial.

20.June 15, 2018: PET/CT scan after 3 months on pembrolizumab reveal a **complete radiographic response** to therapy.

35 yo premenopausal female with a BRCA mutation, diagnosed with node-positive, ER+, HER2+ right breast cancer. Treated with neoadjuvant dual HER2-directed therapy, followed by AC, with residual disease class I. Metastatic disease to the brain noted 4 months after surgery, treated with gamma knife, followed by ado-trastuzumab emtansine. Liver metastasis noted, treatment changed to capecitabine/lapatinib, with subsequent progression, and biopsy revealing PDL-1 positive disease, currently on pembrolizumab leading to a complete response.

Oncology history snapshot: [bolded paragraph above]

Date of diagnosis: Aug 2016 (operable); Sep 2017 (metastatic)

Sites of involvement: brain, liver

Molecular features: ER+, PR+, HER2+, PD-L1+

Prior therapies:

- **early-stage:** THP-ddAC (residual disease)
- **metastatic:** T-DM1, pembro (on trial)
- **procedures:** gamma knife to brain mets